

Table 7: Zero-shot accuracy for direct answering and CoT prompts on all datasets

Dataset	Type	Model	zero-shot CoT accuracy	zero-shot DA accuracy
StrategyQA	Commonsense	GPT-4o Mini	82.2	84.5
StrategyQA	Commonsense	Gpt-4o	84.5	85.5
StrategyQA	Commonsense	Claude-3 Haiku	73.4	65.0
StrategyQA	Commonsense	Claude-3.5 Sonnet	80.1	76.3
StrategyQA	Commonsense	Gemini 1.5 Flash	72.5	75.2
StrategyQA	Commonsense	Gemini 1.5 Pro	74.0	71.4
MuSR Object Placements	Soft Reasoning	Llama 2 7b	36.3	30.5
MuSR Object Placements	Soft Reasoning	Mistral 7b	50.8	43.4
MuSR Object Placements	Soft Reasoning	Llama 3.1 8b	55.5	53.5
MuSR Object Placements	Soft Reasoning	Llama 3.1 70b	65.6	43.8
MuSR Object Placements	Soft Reasoning	Gemma 2 9b	63.3	57.0
MuSR Object Placements	Soft Reasoning	Phi-3 Small 8k	53.1	55.1
MuSR Object Placements	Soft Reasoning	Qwen 2 7b	48.8	48.4
MuSR Object Placements	Soft Reasoning	Qwen 2 72b	61.7	45.7
MuSR Object Placements	Soft Reasoning	GPT-4o Mini	59.0	55.0
MuSR Object Placements	Soft Reasoning	Gpt-4o	67.6	45.3
MuSR Object Placements	Soft Reasoning	Claude-3 Haiku	46.9	52.3
MuSR Object Placements	Soft Reasoning	Claude-3.5 Sonnet	69.5	51.2
MuSR Object Placements	Soft Reasoning	Gemini 1.5 Flash	61.7	56.2
MuSR Object Placements	Soft Reasoning	Gemini 1.5 Pro	66.4	50.0
FOLIO	Symbolic	Llama 2 7b	36.5	33.0
FOLIO	Symbolic	Mistral 7b	50.7	41.9
FOLIO	Symbolic	Llama 3.1 8b	58.6	56.7
FOLIO	Symbolic	Llama 3.1 70b	70.9	69.0
FOLIO	Symbolic	Gemma 2 9b	66.0	55.7
FOLIO	Symbolic	Phi-3 Small 8k	68.0	59.6
FOLIO	Symbolic	Qwen 2 7b	60.6	51.2
FOLIO	Symbolic	Qwen 2 72b	65.0	65.0
FOLIO	Symbolic	GPT-4o Mini	65.0	58.1
FOLIO	Symbolic	Gpt-4o	79.8	62.6
FOLIO	Symbolic	Claude-3 Haiku	61.6	48.8
FOLIO	Symbolic	Claude-3.5 Sonnet	73.9	68.5
FOLIO	Symbolic	Gemini 1.5 Flash	74.9	69.5
FOLIO	Symbolic	Gemini 1.5 Pro	73.9	74.4
ContextHub Deductive L2	Symbolic	Llama 2 7b	34.8	12.6
ContextHub Deductive L2	Symbolic	Mistral 7b	48.8	55.1
ContextHub Deductive L2	Symbolic	Llama 3.1 8b	52.8	21.5
ContextHub Deductive L2	Symbolic	Llama 3.1 70b	50.0	41.1
ContextHub Deductive L2	Symbolic	Gemma 2 9b	50.0	43.0
ContextHub Deductive L2	Symbolic	Phi-3 Small 8k	52.4	49.1
ContextHub Deductive L2	Symbolic	Qwen 2 7b	51.3	39.8
ContextHub Deductive L2	Symbolic	Qwen 2 72b	52.8	44.0
ContextHub Deductive L2	Symbolic	GPT-4o Mini	47.0	42.0
ContextHub Deductive L2	Symbolic	Gpt-4o	54.5	45.6
ContextHub Deductive L2	Symbolic	Claude-3 Haiku	45.2	41.8
ContextHub Deductive L2	Symbolic	Claude-3.5 Sonnet	53.0	46.2
ContextHub Deductive L2	Symbolic	Gemini 1.5 Flash	45.0	39.5
ContextHub Deductive L2	Symbolic	Gemini 1.5 Pro	57.3	43.3
ContextHub Abductive L2	Symbolic	Llama 2 7b	34.3	31.9
ContextHub Abductive L2	Symbolic	Mistral 7b	34.0	25.7
ContextHub Abductive L2	Symbolic	Llama 3.1 8b	41.3	37.3
ContextHub Abductive L2	Symbolic	Llama 3.1 70b	51.0	44.4
ContextHub Abductive L2	Symbolic	Gemma 2 9b	41.5	32.9
ContextHub Abductive L2	Symbolic	Phi-3 Small 8k	44.3	32.8
ContextHub Abductive L2	Symbolic	Qwen 2 7b	37.8	33.4
ContextHub Abductive L2	Symbolic	Qwen 2 72b	45.5	32.2
ContextHub Abductive L2	Symbolic	GPT-4o Mini	65.0	55.0
ContextHub Abductive L2	Symbolic	Gpt-4o	57.5	46.8
ContextHub Abductive L2	Symbolic	Claude-3 Haiku	37.0	31.4
ContextHub Abductive L2	Symbolic	Claude-3.5 Sonnet	56.8	40.4
ContextHub Abductive L2	Symbolic	Gemini 1.5 Flash	53.1	32.2
ContextHub Abductive L2	Symbolic	Gemini 1.5 Pro	53.5	43.7
MMLU Pro	Knowledge	Llama 2 7b	19.9	19.6
MMLU Pro	Knowledge	Mistral 7b	31.6	28.4
MMLU Pro	Knowledge	Llama 3.1 8b	44.8	38.0
MMLU Pro	Knowledge	Llama 3.1 70b	64.9	55.0
MMLU Pro	Knowledge	Gemma 2 9b	48.1	42.7
MMLU Pro	Knowledge	Phi-3 Small 8k	54.8	43.7
MMLU Pro	Knowledge	Qwen 2 7b	45.0	36.2
MMLU Pro	Knowledge	Qwen 2 72b	62.8	44.3
MMLU Pro	Knowledge	GPT-4o Mini	62.3	42.6
MMLU Pro	Knowledge	Gpt-4o	72.1	55.0
MMLU Pro	Knowledge	Claude-3 Haiku	47.6	39.0
MMLU Pro	Knowledge	Claude-3.5 Sonnet	73.4	57.2
MMLU Pro	Knowledge	Gemini 1.5 Flash	58.5	47.2
MMLU Pro	Knowledge	Gemini 1.5 Pro	65.3	57.4
MuSR Murder Mysteries	Soft Reasoning	Llama 2 7b	50.0	50.0

Table 7: Zero-shot accuracy for direct answering and CoT prompts on all datasets

Dataset	Type	Model	zero-shot CoT accuracy	zero-shot DA accuracy
MuSR Murder Mysteries	Soft Reasoning	Mistral 7b	62.8	55.6
MuSR Murder Mysteries	Soft Reasoning	Llama 3.1 8b	70.4	57.2
MuSR Murder Mysteries	Soft Reasoning	Llama 3.1 70b	73.6	69.6
MuSR Murder Mysteries	Soft Reasoning	Gemma 2 9b	76.8	61.6
MuSR Murder Mysteries	Soft Reasoning	Phi-3 Small 8k	61.6	58.8
MuSR Murder Mysteries	Soft Reasoning	Qwen 2 7b	59.2	53.2
MuSR Murder Mysteries	Soft Reasoning	Qwen 2 72b	80.8	64.4
MuSR Murder Mysteries	Soft Reasoning	GPT-4o Mini	71.2	63.6
MuSR Murder Mysteries	Soft Reasoning	Gpt-4o	87.6	70.8
MuSR Murder Mysteries	Soft Reasoning	Claude-3 Haiku	62.4	56.8
MuSR Murder Mysteries	Soft Reasoning	Claude-3.5 Sonnet	85.2	70.4
MuSR Murder Mysteries	Soft Reasoning	Gemini 1.5 Flash	70.8	58.4
MuSR Murder Mysteries	Soft Reasoning	Gemini 1.5 Pro	77.6	64.0
ContextHub Deductive L1	Symbolic	Llama 2 7b	47.7	8.3
ContextHub Deductive L1	Symbolic	Mistral 7b	50.3	67.3
ContextHub Deductive L1	Symbolic	Llama 3.1 8b	50.7	23.3
ContextHub Deductive L1	Symbolic	Llama 3.1 70b	53.8	40.7
ContextHub Deductive L1	Symbolic	Gemma 2 9b	56.3	39.2
ContextHub Deductive L1	Symbolic	Phi-3 Small 8k	54.8	50.2
ContextHub Deductive L1	Symbolic	Qwen 2 7b	59.3	43.3
ContextHub Deductive L1	Symbolic	Qwen 2 72b	51.5	44.0
ContextHub Deductive L1	Symbolic	GPT-4o Mini	49.3	41.5
ContextHub Deductive L1	Symbolic	Gpt-4o	59.3	49.0
ContextHub Deductive L1	Symbolic	Claude-3 Haiku	50.5	39.7
ContextHub Deductive L1	Symbolic	Claude-3.5 Sonnet	54.5	47.0
ContextHub Deductive L1	Symbolic	Gemini 1.5 Flash	47.3	38.5
ContextHub Deductive L1	Symbolic	Gemini 1.5 Pro	57.3	46.0
ContextHub Abductive L1	Symbolic	Llama 2 7b	29.4	16.4
ContextHub Abductive L1	Symbolic	Mistral 7b	46.9	25.8
ContextHub Abductive L1	Symbolic	Llama 3.1 8b	43.6	24.2
ContextHub Abductive L1	Symbolic	Llama 3.1 70b	55.3	43.9
ContextHub Abductive L1	Symbolic	Gemma 2 9b	61.9	58.9
ContextHub Abductive L1	Symbolic	Phi-3 Small 8k	62.5	60.3
ContextHub Abductive L1	Symbolic	Qwen 2 7b	52.2	47.5
ContextHub Abductive L1	Symbolic	Qwen 2 72b	61.9	45.0
ContextHub Abductive L1	Symbolic	GPT-4o Mini	61.1	42.2
ContextHub Abductive L1	Symbolic	Gpt-4o	74.2	65.6
ContextHub Abductive L1	Symbolic	Claude-3 Haiku	35.3	22.8
ContextHub Abductive L1	Symbolic	Claude-3.5 Sonnet	80.8	60.3
ContextHub Abductive L1	Symbolic	Gemini 1.5 Flash	66.4	47.2
ContextHub Abductive L1	Symbolic	Gemini 1.5 Pro	62.2	60.0
Big-Bench Hard	Symbolic	Llama 2 7b	29.8	31.9
Big-Bench Hard	Symbolic	Mistral 7b	39.3	35.1
Big-Bench Hard	Symbolic	Llama 3.1 8b	62.8	45.6
Big-Bench Hard	Symbolic	Llama 3.1 70b	78.9	54.8
Big-Bench Hard	Symbolic	Gemma 2 9b	58.7	50.8
Big-Bench Hard	Symbolic	Phi-3 Small 8k	70.0	55.1
Big-Bench Hard	Symbolic	Qwen 2 7b	52.6	47.6
Big-Bench Hard	Symbolic	Qwen 2 72b	75.1	59.0
Big-Bench Hard	Symbolic	GPT-4o Mini	77.7	49.7
Big-Bench Hard	Symbolic	Gpt-4o	84.6	64.5
Big-Bench Hard	Symbolic	Claude-3 Haiku	62.4	47.3
Big-Bench Hard	Symbolic	Claude-3.5 Sonnet	83.6	56.9
Big-Bench Hard	Symbolic	Gemini 1.5 Flash	71.3	55.4
Big-Bench Hard	Symbolic	Gemini 1.5 Pro	71.6	50.3
MATH	Mathematical	Llama 2 7b	4.2	4.0
MATH	Mathematical	Mistral 7b	12.4	6.1
MATH	Mathematical	Llama 3.1 8b	47.2	13.8
MATH	Mathematical	Llama 3.1 70b	64.4	22.8
MATH	Mathematical	Gemma 2 9b	45.6	19.1
MATH	Mathematical	Phi-3 Small 8k	43.2	18.5
MATH	Mathematical	Qwen 2 7b	53.7	13.3
MATH	Mathematical	Qwen 2 72b	63.5	23.8
MATH	Mathematical	GPT-4o Mini	69.6	24.3
MATH	Mathematical	Gpt-4o	73.3	35.2
MATH	Mathematical	Claude-3 Haiku	32.7	17.4
MATH	Mathematical	Claude-3.5 Sonnet	63.8	34.6
MATH	Mathematical	Gemini 1.5 Flash	54.5	31.3
MATH	Mathematical	Gemini 1.5 Pro	62.1	39.4
GSM8k-Hard	Mathematical	Llama 2 7b	6.7	1.8
GSM8k-Hard	Mathematical	Mistral 7b	21.0	3.0
GSM8k-Hard	Mathematical	Llama 3.1 8b	34.4	6.0
GSM8k-Hard	Mathematical	Llama 3.1 70b	46.6	14.0
GSM8k-Hard	Mathematical	Gemma 2 9b	40.9	8.8
GSM8k-Hard	Mathematical	Phi-3 Small 8k	33.0	6.9
GSM8k-Hard	Mathematical	Qwen 2 7b	48.4	5.0
GSM8k-Hard	Mathematical	Qwen 2 72b	54.8	13.7

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Dataset	Type	Model	zero-shot CoT accuracy	zero-shot DA accuracy
GSM8k-Hard	Mathematical	GPT-4o Mini	53.9	11.7
GSM8k-Hard	Mathematical	Gpt-4o	60.3	26.0
GSM8k-Hard	Mathematical	Claude-3 Haiku	45.3	9.6
GSM8k-Hard	Mathematical	Claude-3.5 Sonnet	50.8	32.3
GSM8k-Hard	Mathematical	Gemini 1.5 Flash	54.6	16.2
GSM8k-Hard	Mathematical	Gemini 1.5 Pro	58.2	26.2
GSM8k	Mathematical	Llama 2 7b	29.6	6.9
GSM8k	Mathematical	Mistral 7b	59.2	10.2
GSM8k	Mathematical	Llama 3.1 8b	85.4	18.5
GSM8k	Mathematical	Llama 3.1 70b	85.6	37.0
GSM8k	Mathematical	Gemma 2 9b	89.2	24.9
GSM8k	Mathematical	Phi-3 Small 8k	90.0	24.9
GSM8k	Mathematical	Qwen 2 7b	87.9	20.7
GSM8k	Mathematical	Qwen 2 72b	94.6	40.1
GSM8k	Mathematical	GPT-4o Mini	94.1	31.8
GSM8k	Mathematical	Gpt-4o	95.8	58.8
GSM8k	Mathematical	Claude-3 Haiku	89.4	22.9
GSM8k	Mathematical	Claude-3.5 Sonnet	96.1	62.2
GSM8k	Mathematical	Gemini 1.5 Flash	91.4	38.6
GSM8k	Mathematical	Gemini 1.5 Pro	92.7	52.4

F.2 FULL FEW-SHOT RESULTS

Table 8: Few-shot accuracy for direct answering and CoT prompts on all datasets

Dataset	Type	Model	few-shot CoT accuracy	few-shot DA accuracy
AGIEval LSAT RC	Soft Reasoning	Llama 2 7b	33.1	38.7
AGIEval LSAT RC	Soft Reasoning	Mistral 7b	52.4	57.2
AGIEval LSAT RC	Soft Reasoning	Llama 3.1 8b	60.2	70.3
AGIEval LSAT RC	Soft Reasoning	Llama 3.1 70b	84.4	88.8
AGIEval LSAT RC	Soft Reasoning	Gemma 2 9b	74.3	79.2
AGIEval LSAT RC	Soft Reasoning	Phi-3 Small 8k	63.2	65.1
AGIEval LSAT RC	Soft Reasoning	Qwen 2 7b	61.7	68.8
AGIEval LSAT RC	Soft Reasoning	Qwen 2 72b	85.9	85.9
AGIEval LSAT RC	Soft Reasoning	GPT-4o Mini	77.3	71.4
AGIEval LSAT RC	Soft Reasoning	Gemini 1.5 Flash	79.2	81.8
AGIEval LSAT LR	Soft Reasoning	Llama 2 7b	33.7	34.7
AGIEval LSAT LR	Soft Reasoning	Mistral 7b	46.1	48.0
AGIEval LSAT LR	Soft Reasoning	Llama 3.1 8b	55.7	58.0
AGIEval LSAT LR	Soft Reasoning	Llama 3.1 70b	83.3	85.1
AGIEval LSAT LR	Soft Reasoning	Gemma 2 9b	65.7	68.2
AGIEval LSAT LR	Soft Reasoning	Phi-3 Small 8k	64.7	59.2
AGIEval LSAT LR	Soft Reasoning	Qwen 2 7b	54.1	61.2
AGIEval LSAT LR	Soft Reasoning	Qwen 2 72b	77.5	79.6
AGIEval LSAT LR	Soft Reasoning	GPT-4o Mini	68.4	64.5
AGIEval LSAT LR	Soft Reasoning	Gemini 1.5 Flash	68.6	72.9
GPQA	Mathematical	Mistral 7b	23.0	25.9
GPQA	Mathematical	Llama 3.1 8b	22.1	27.2
GPQA	Mathematical	Llama 3.1 70b	24.8	24.3
GPQA	Mathematical	Gemma 2 9b	19.9	22.3
GPQA	Mathematical	Phi-3 Small 8k	23.9	22.5
GPQA	Mathematical	Qwen 2 7b	23.4	21.2
GPQA	Mathematical	Qwen 2 72b	22.8	19.9
GPQA	Mathematical	GPT-4o Mini	20.0	20.0
GPQA	Mathematical	Gemini 1.5 Flash	21.9	24.6
CommonsenseQA	Commonsense	Llama 2 7b	18.2	19.2
CommonsenseQA	Commonsense	Mistral 7b	73.6	70.4
CommonsenseQA	Commonsense	Llama 3.1 8b	74.0	76.5
CommonsenseQA	Commonsense	Llama 3.1 70b	84.7	84.6
CommonsenseQA	Commonsense	Gemma 2 9b	81.8	80.8
CommonsenseQA	Commonsense	Phi-3 Small 8k	80.8	80.4
CommonsenseQA	Commonsense	Qwen 2 7b	80.3	72.9
CommonsenseQA	Commonsense	Qwen 2 72b	88.4	87.8
CommonsenseQA	Commonsense	GPT-4o Mini	84.7	84.7
CommonsenseQA	Commonsense	Gemini 1.5 Flash	81.7	83.3
AGIEval LSAT AR	Soft Reasoning	Llama 2 7b	19.6	18.7
AGIEval LSAT AR	Soft Reasoning	Mistral 7b	20.9	22.6
AGIEval LSAT AR	Soft Reasoning	Llama 3.1 8b	24.8	26.1
AGIEval LSAT AR	Soft Reasoning	Llama 3.1 70b	36.1	30.9
AGIEval LSAT AR	Soft Reasoning	Gemma 2 9b	22.2	28.7
AGIEval LSAT AR	Soft Reasoning	Phi-3 Small 8k	27.8	20.0
AGIEval LSAT AR	Soft Reasoning	Qwen 2 7b	24.3	23.0
AGIEval LSAT AR	Soft Reasoning	Qwen 2 72b	27.0	30.0